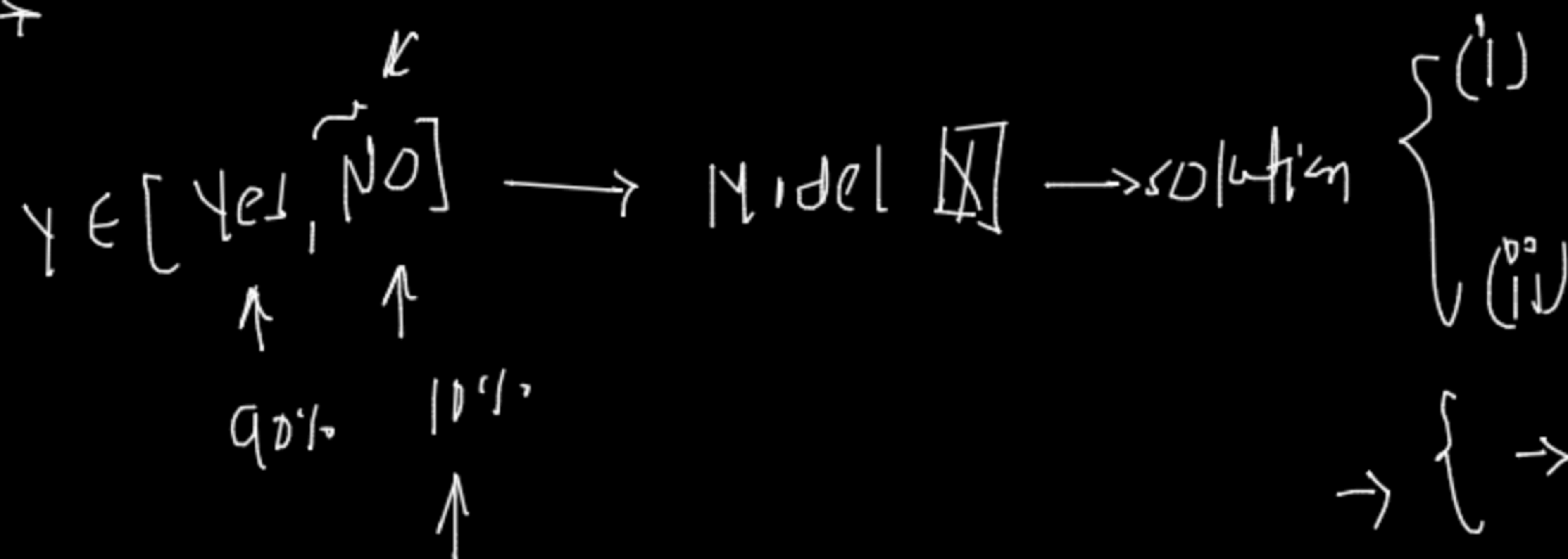


Imbalanced data  $\rightarrow$

classification  $\rightarrow$

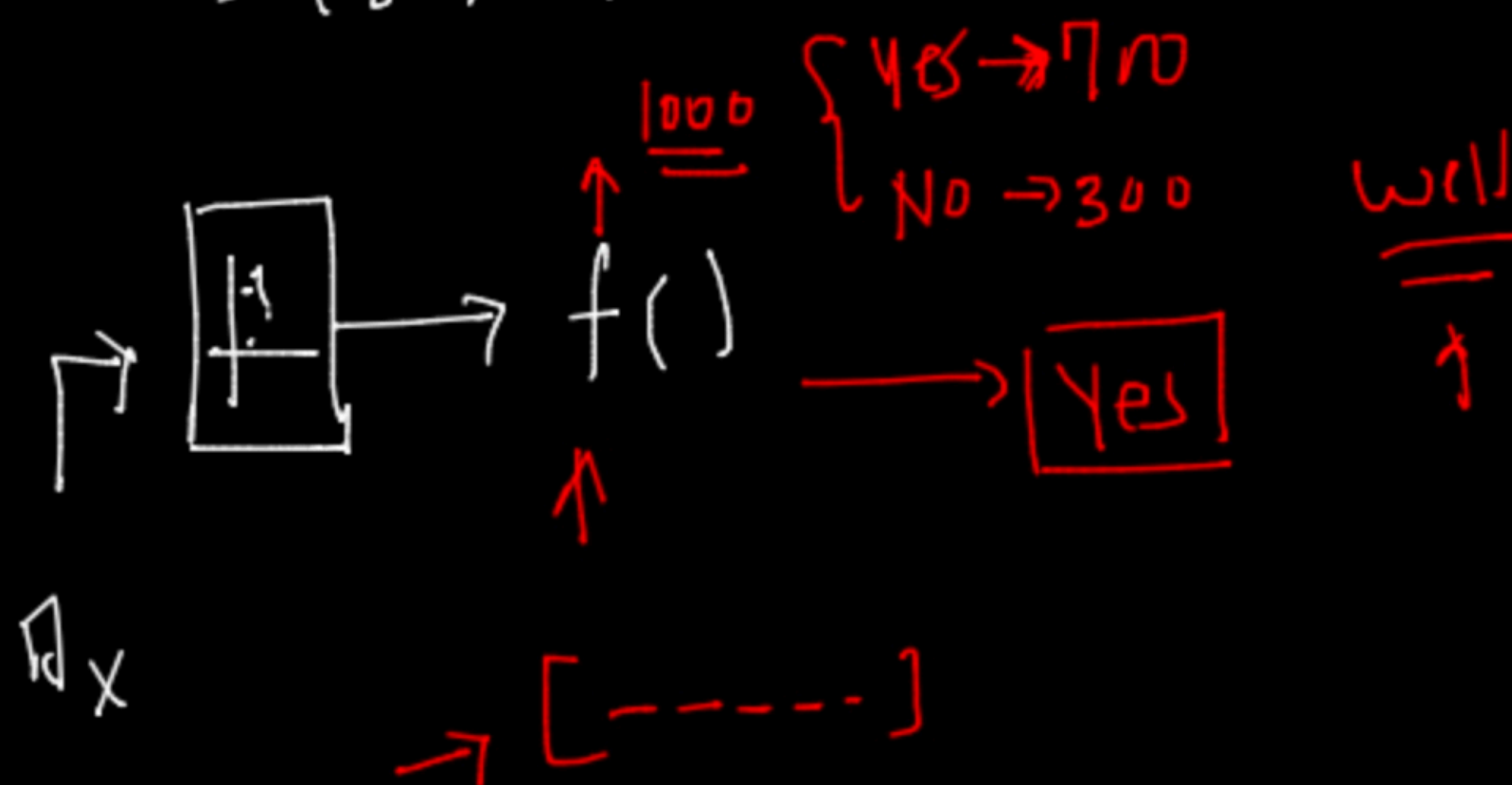
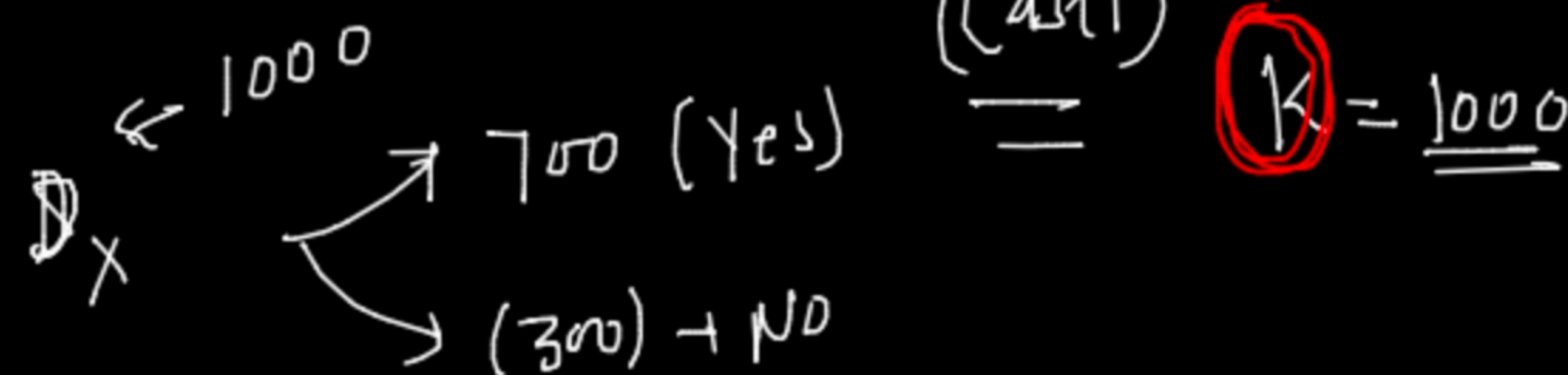


$\rightarrow \{ \rightarrow$  'underfitting'

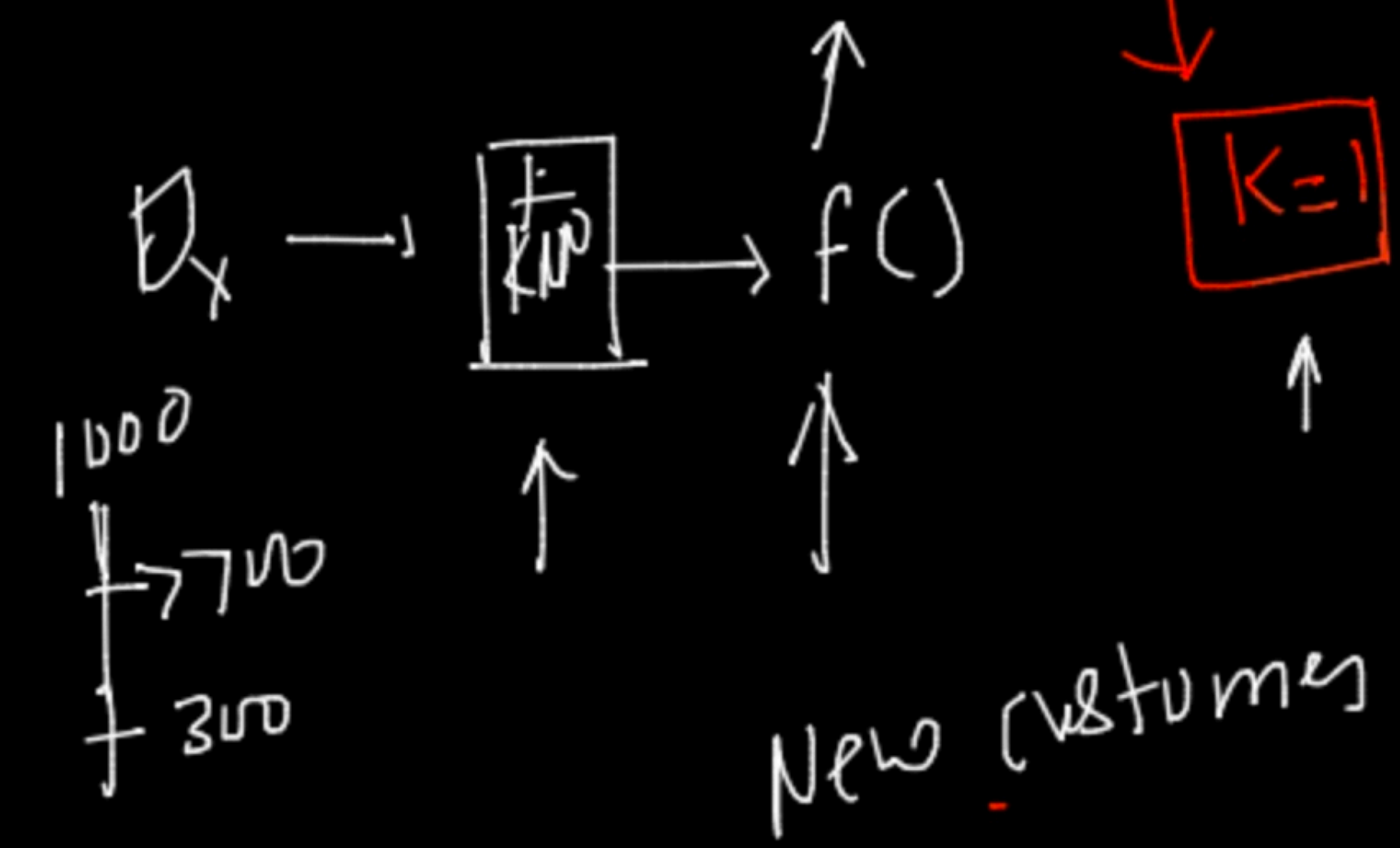
$\Rightarrow$  eg.  $\begin{cases} k \rightarrow k \uparrow ? \\ \uparrow \rightarrow k \downarrow ? \end{cases}$

Best  $\rightarrow$

let



$K=1$



overfitting



fit the model on new data

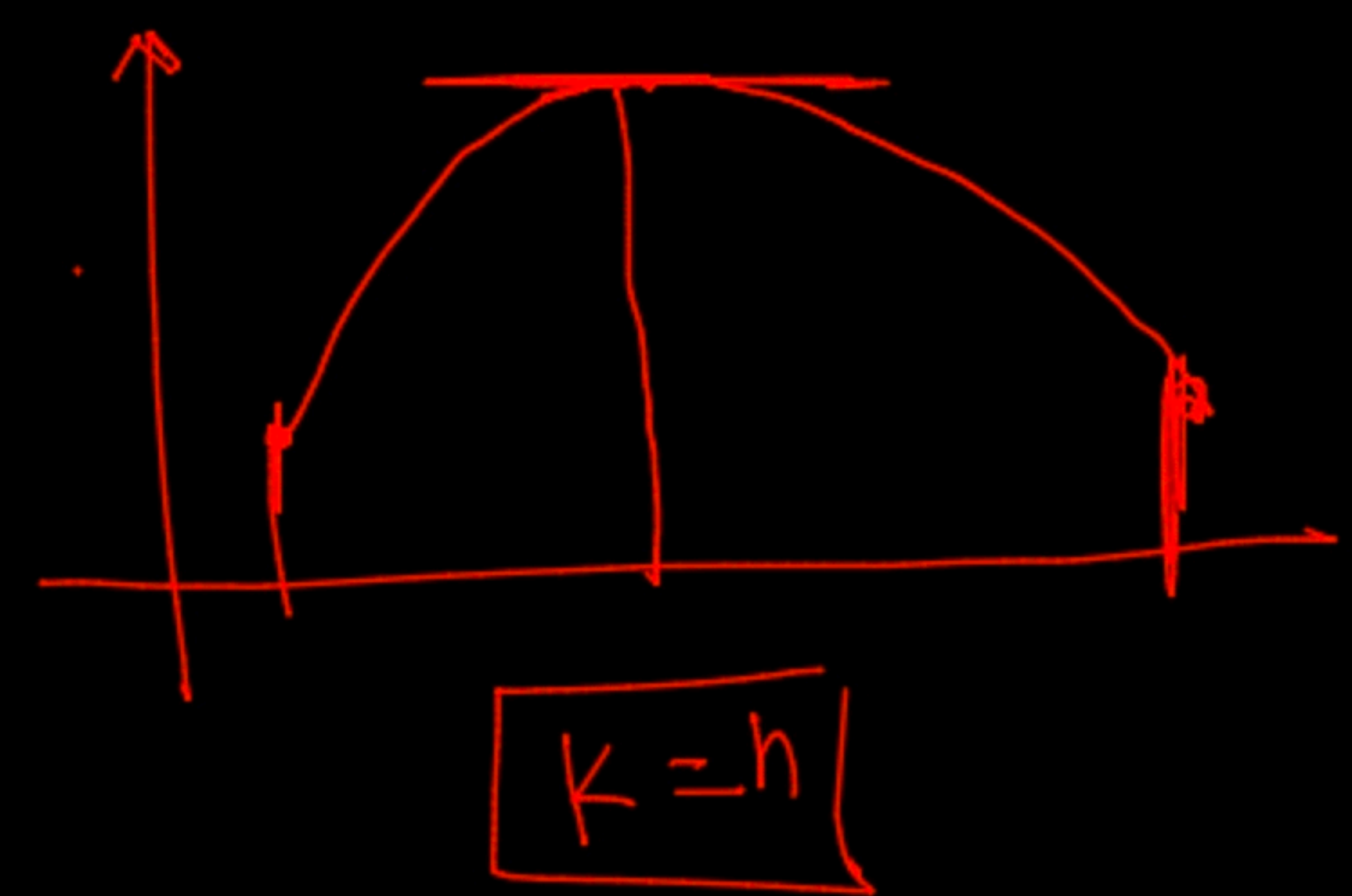
good  $\times$

$K \rightarrow$  control  $L/K$

$\rightarrow$  common  $\rightarrow$  model  $\rightarrow$  Hyper parameters

overfitting and underfitting

$K=1$   $\rightarrow$  error  
 $K=3$   $\rightarrow$  good  
 $K=1000$   $\rightarrow$  error



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KNN

final-accuracy = [ ]

k = [3, 5, 7, 9, 11, 13, 15]

⇒ Multi-class classification → KNN

↳ init →

for i in k:

model = KNN(n=i)

model.fit(X\_train)

⇒ Outlier? (KNN)

(KNN)

y\_pred = model.predict(X\_test)

⇒ impact of scale and col. standardization

acc = accuracy(y\_pred, y\_actual)

final-accuracy.append(acc)

Talking:

Audio

Video

Participants 2

Chat

Share

Pause

Layout

Annotate

Slide control

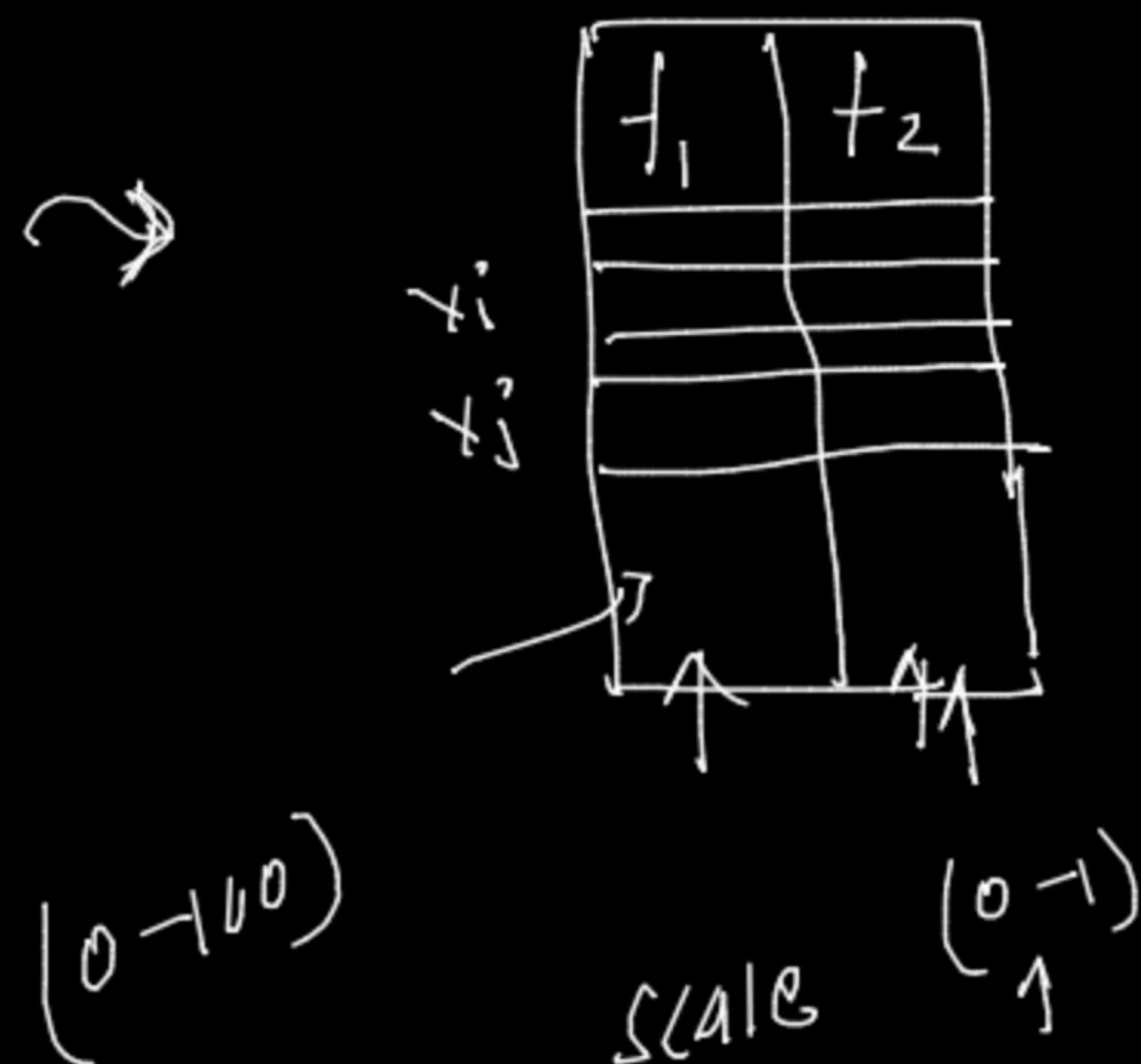
Show meeting

More

You are screen sharing

Stop share

```
final-accuracy = [ ]
k = [3, 5, 7, 9, 11, 13, 15]
for i in k:
    model = KNN(n=i)
    model.fit(X_train)
    y_pred = model.predict(X_test)
    acc = accuracy(y_pred, y_actual)
    final-accuracy.append(acc)
```



↓  
diff

$$x_1 = [23, 0.2]$$

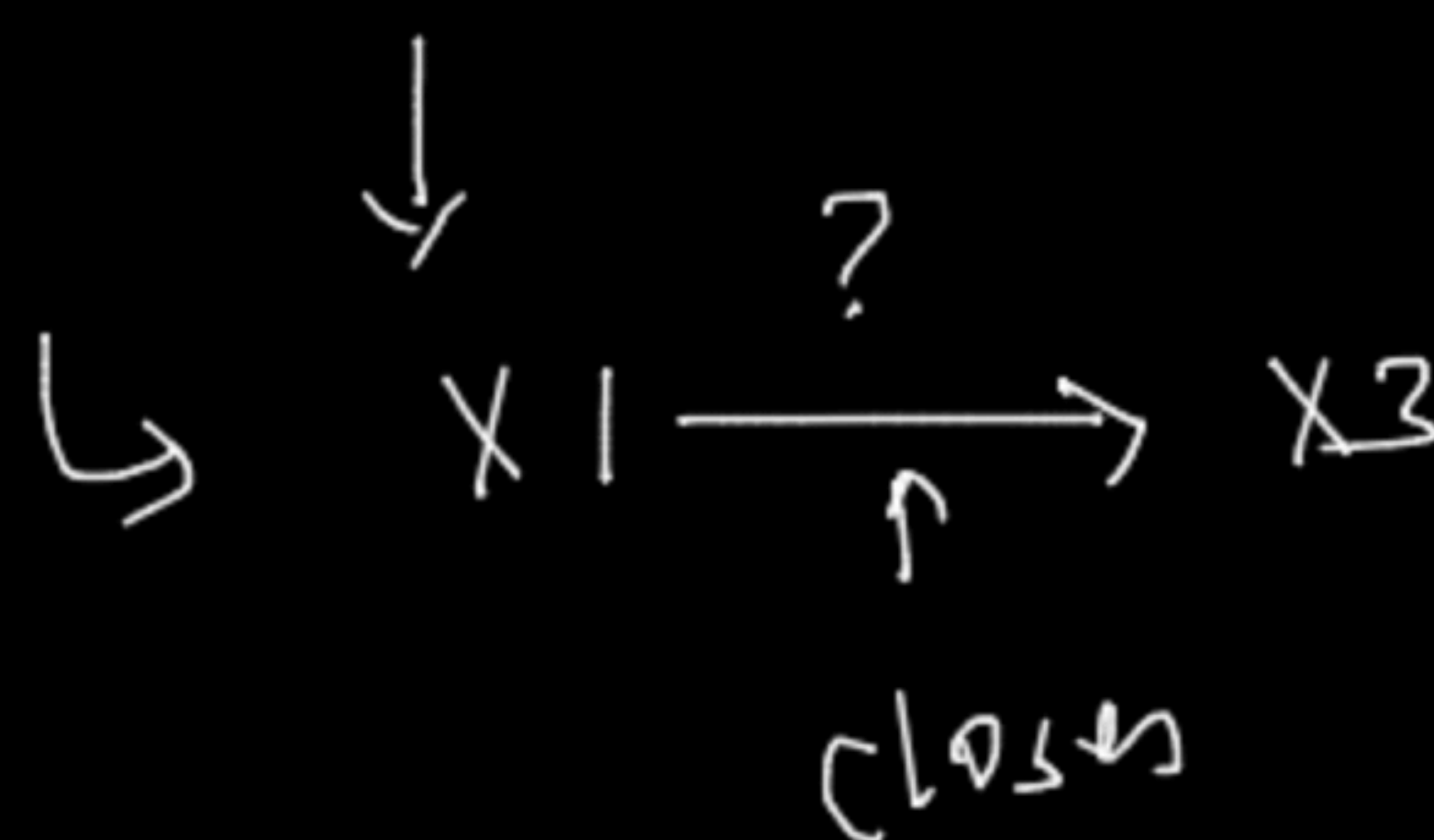
$$x_2 = [28, 0.2]$$

$$x_3 = [23, 1.0]$$

↓ 54m → 80-10

$$\text{distance}(x_1, x_2) = 5$$

$$\text{distance}(x_1, x_3) = 0.8$$



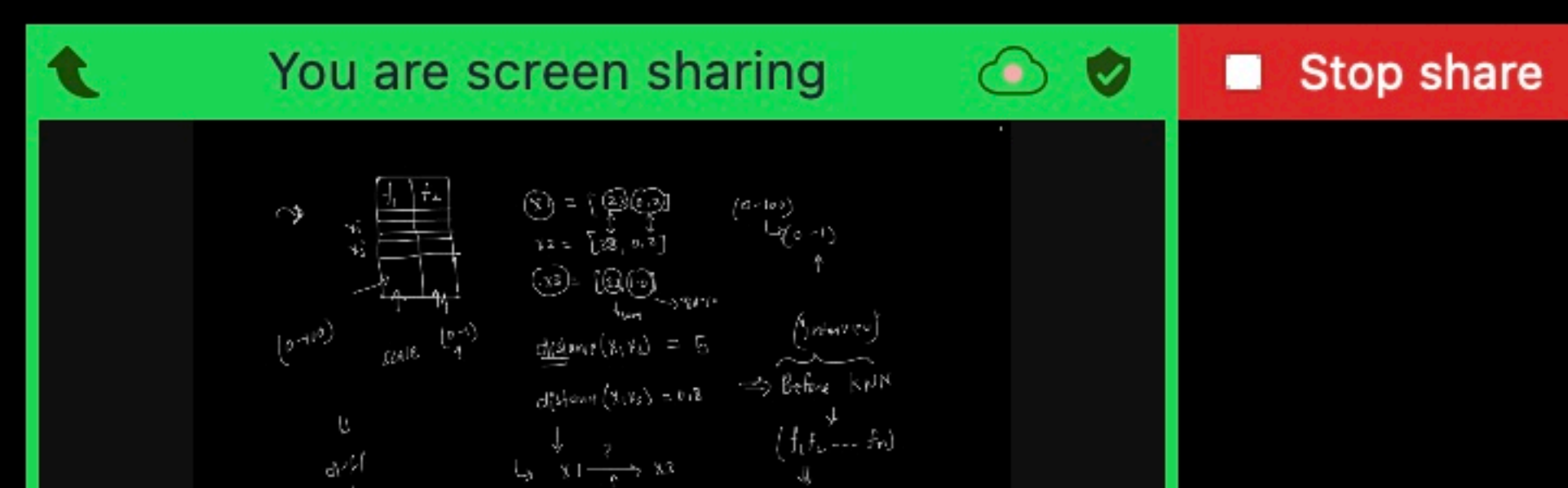
$$(0-100) \rightarrow (0-1)$$

Interview  
⇒ Before KNN

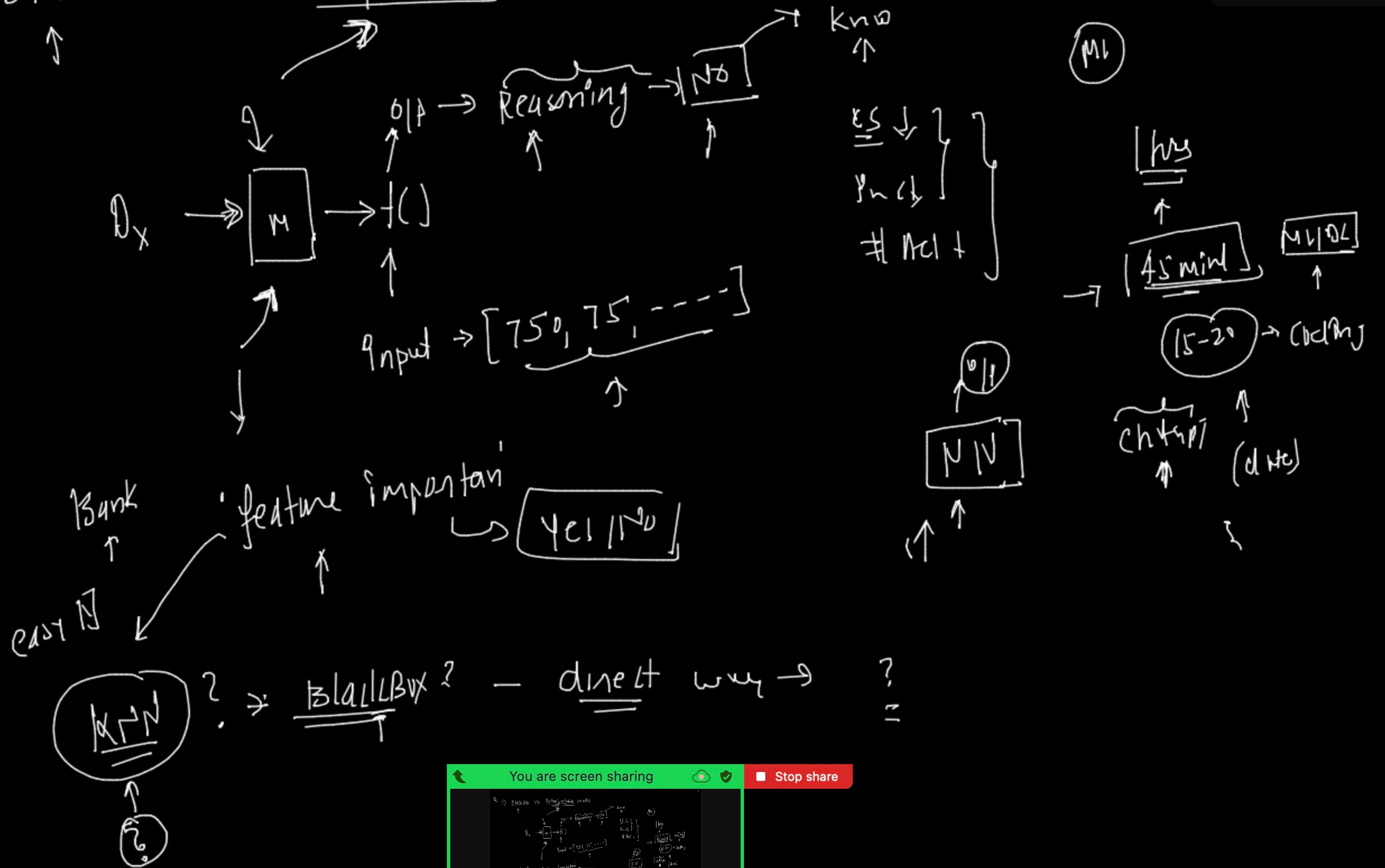
↓  
( $f_1, f_2, \dots, f_n$ )

↓  
col. standardization

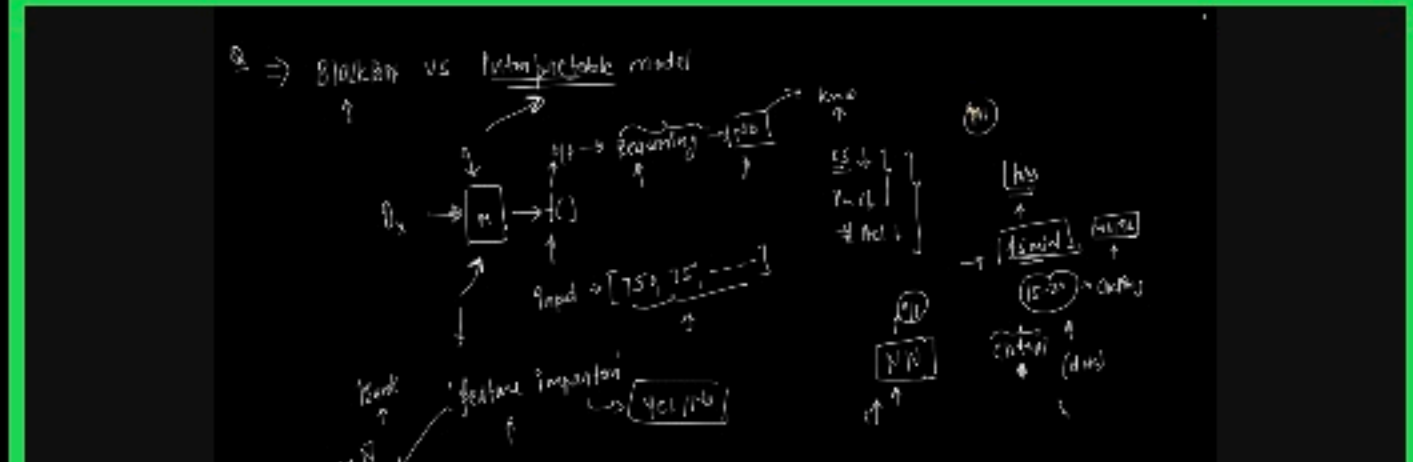
↑  
sklearn.preprocessing



Q  $\Rightarrow$  Black Box vs Interpretable model



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# Feature importance and Forward selection



$(f_1, f_2, f_3, \dots, f_{10})$

RFE

'Sklearn'

12me



ex predict height of person

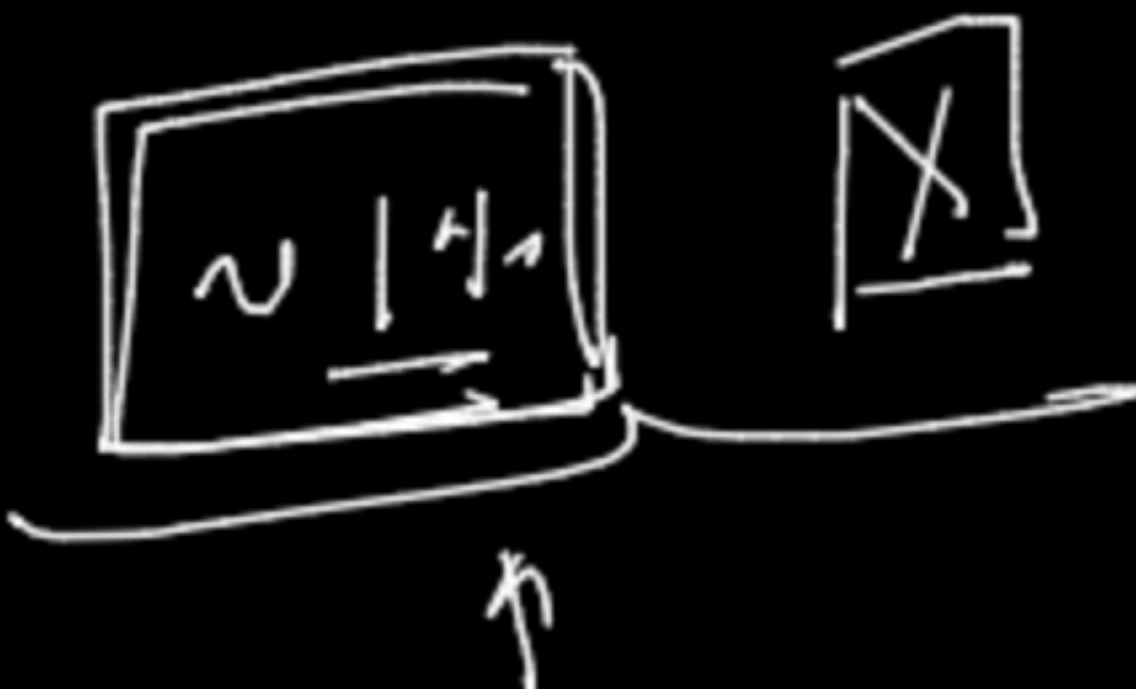
Proof  
Domain

weight, height-color, hair-length, skin-color - - -

3 months

2 months

$D = \{Tr, Te\}$



1 pre-proc { } 94%

Model 1m

Step

$f_1 \rightarrow [MI] \rightarrow a_1$  (All)

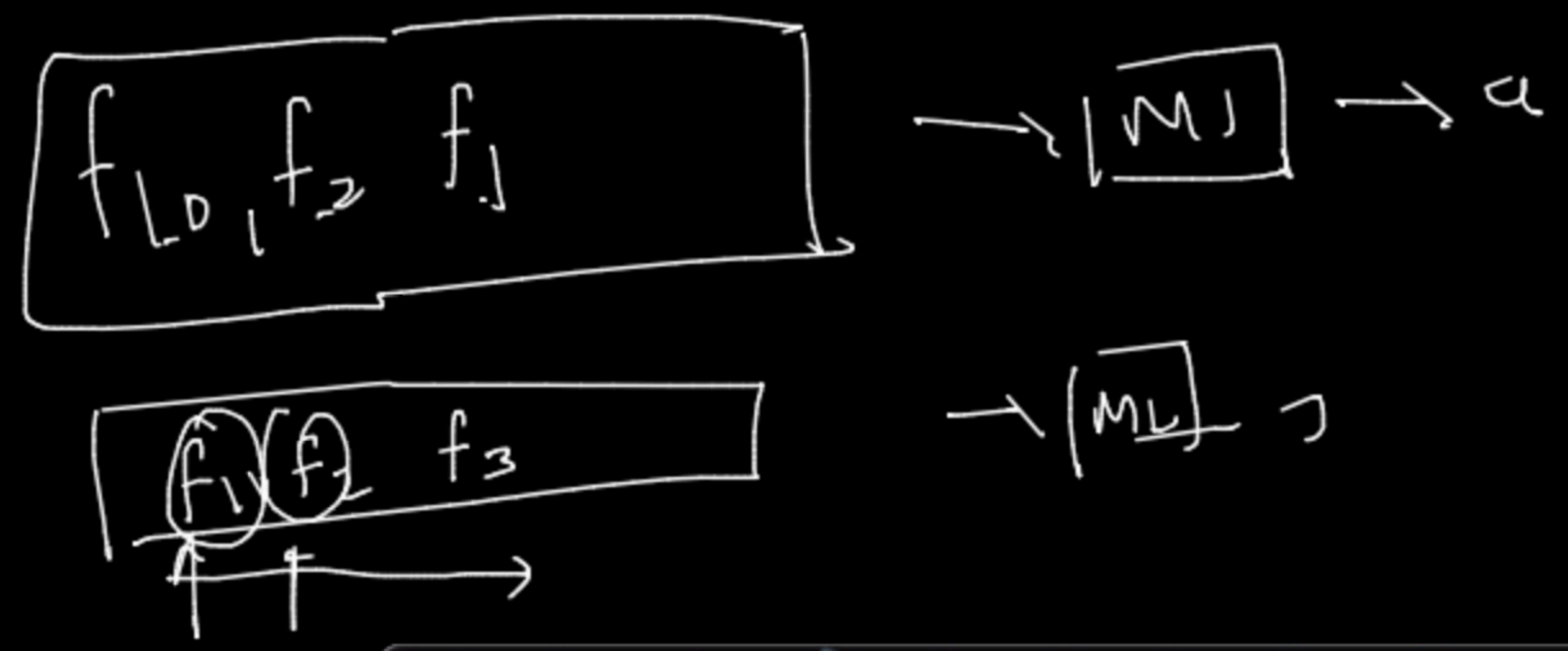
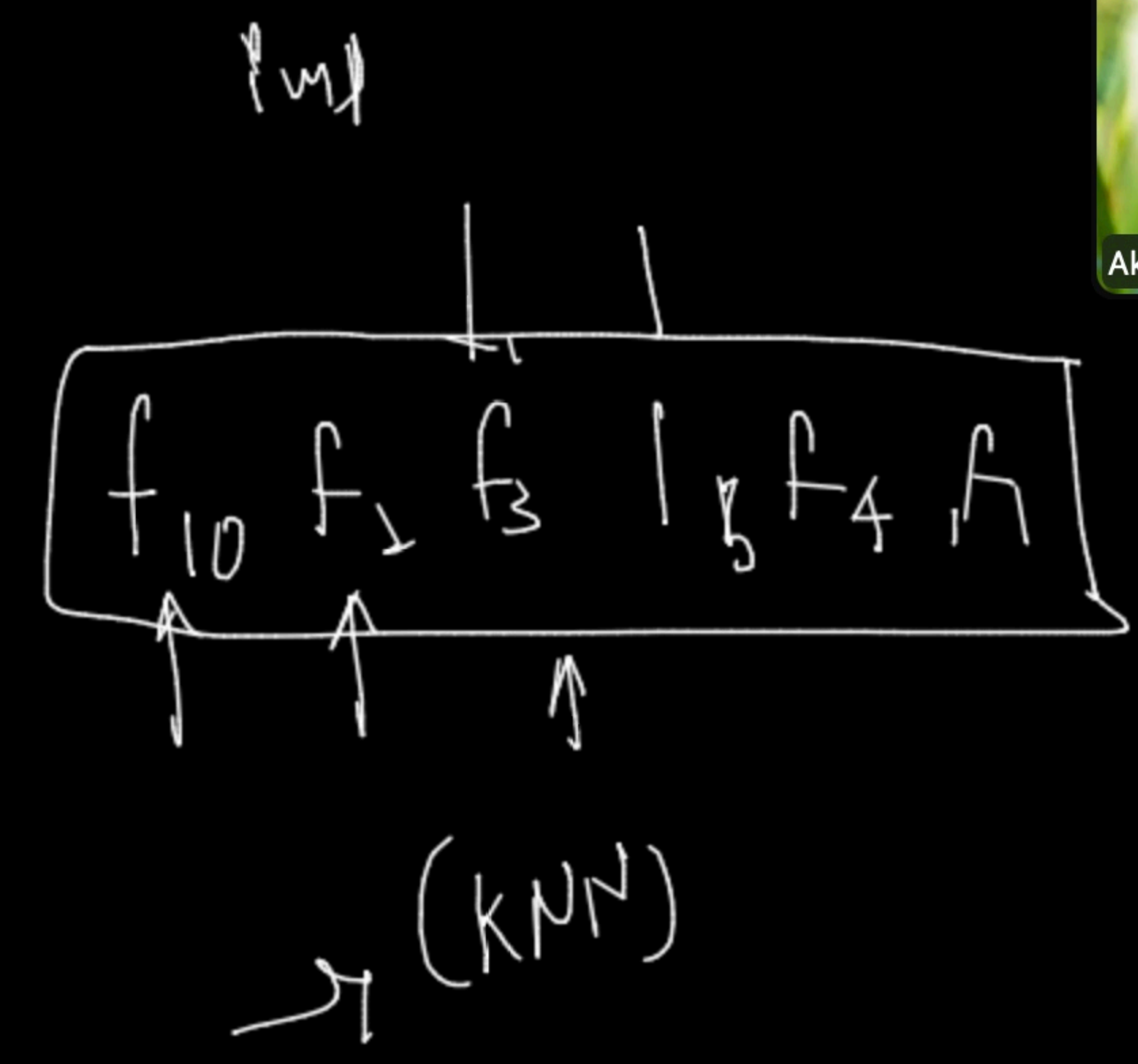
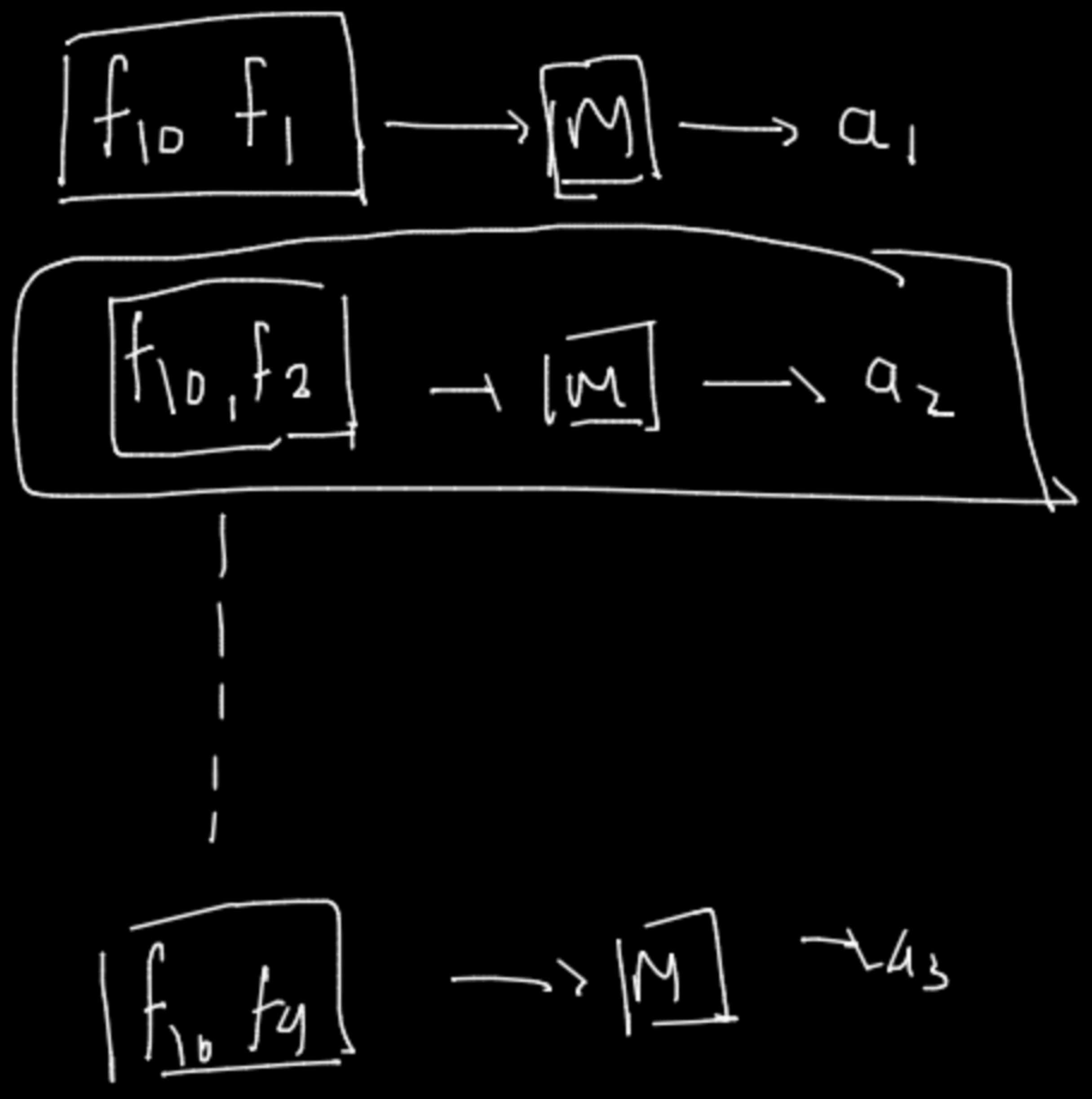
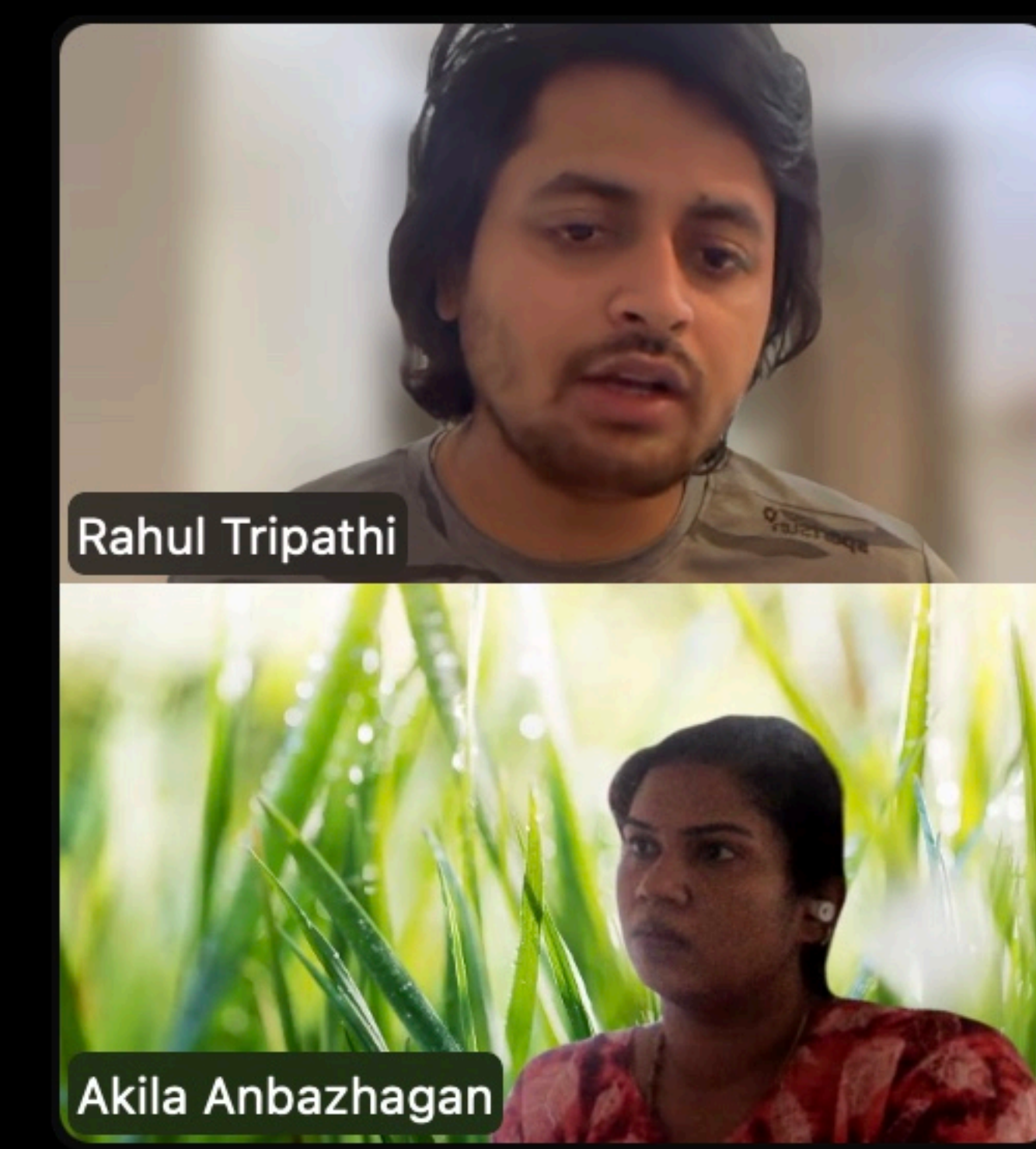
Highest All most imp

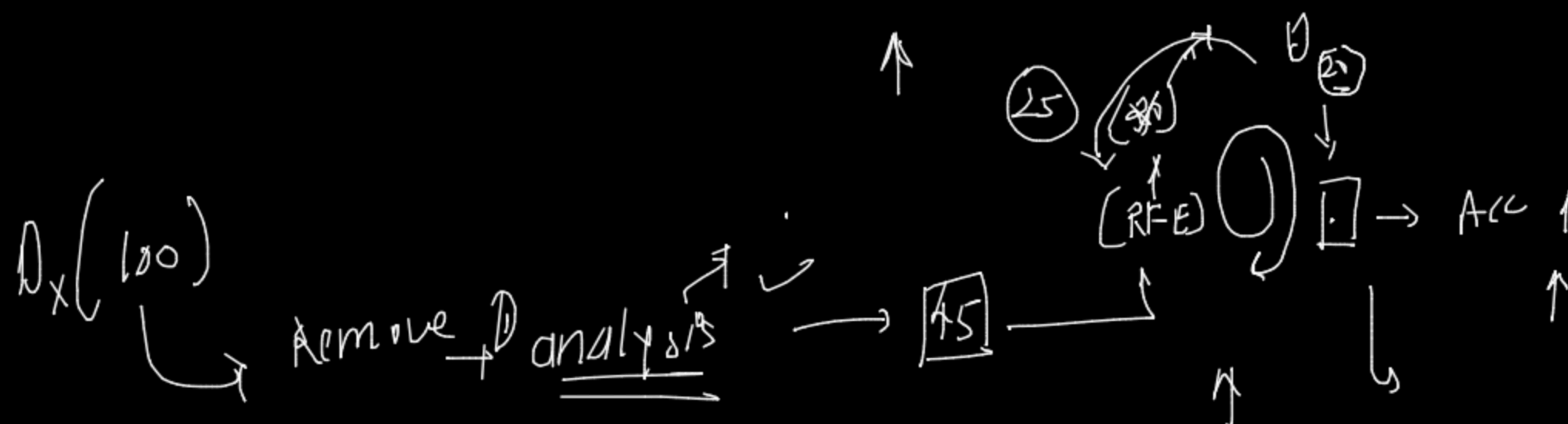
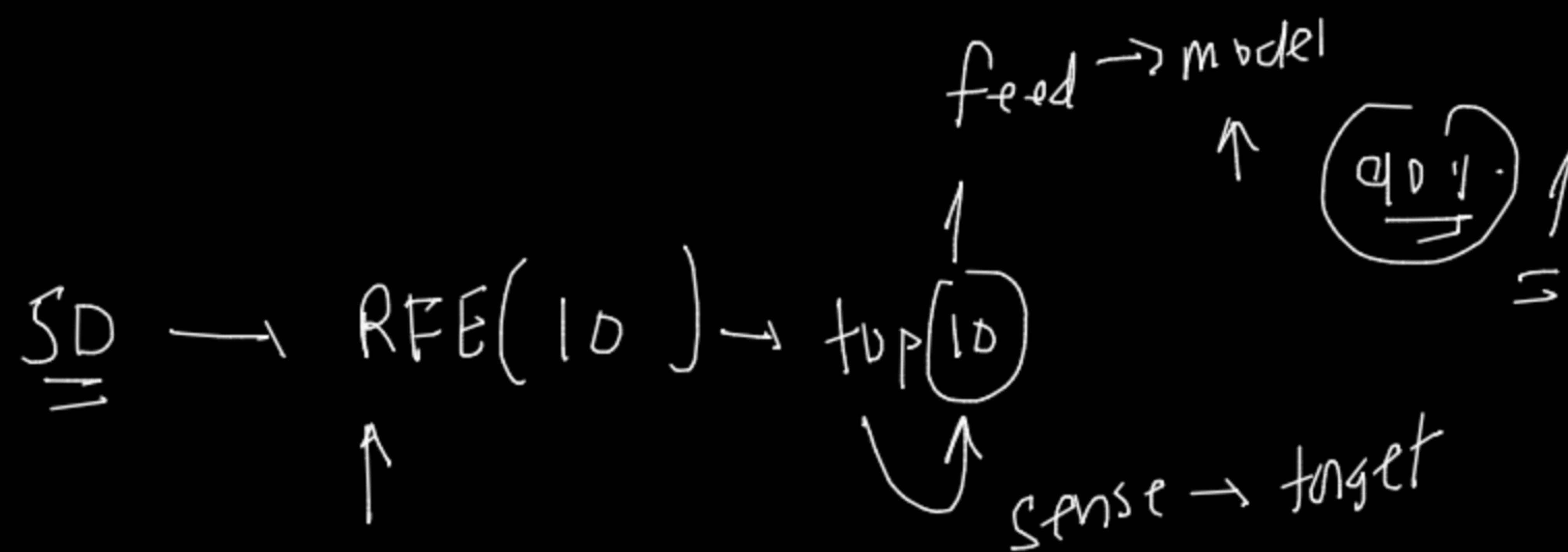
$f_2 \rightarrow [MI] \rightarrow a_2$

$f_{10}$  Best - ?

15% (MI) Engine







→ ② Domain

→ ③ Don't use X

→ ④ Use Keep